

1. A computer algorithm is

1 point

- ☐ A program written in a computer language such as C++, Python, or Perl
- ☒ A precise specification of all the steps needed to compute a solution to a problem
- ☐ A description of the hardware and software capabilities of a computer
- ☐ A protocol for transmitting data over a network

2. You can make a program more efficient by

1 point

- ☒ Re-designing the data structures to require less storage
- ☐ Using a faster computer
- ☐ Running it on the Amazon cloud
- ☐ Replacing your hard drives with faster solid state drives

3. DNA sequences can be represented efficiently using

1 point

- ☒ Two bits for each of the 4 possible bases
- ☐ One byte for each of the 4 possible bases
- ☐ The compressed BAM file format
- ☐ The SAM alignment format

4. A programming language is

1 point

- ☒ A formal language used to instruct computers what to do
- ☐ The way we describe high-level algorithms
- ☐ Anything written in Python
- ☐ A method for translating between languages such as English and French

5. Software engineering involves

1 point

- ☐ Updating code so that it remains compatible with other software systems.
- ☐ Testing programs on a wide range of examples to see if they perform as expected
- ☒ All of these options
- ☐ Debugging computer code

6. Bowtie, TopHat, and Cufflinks are

1 point

- ☒ Programs for analysis of RNA-seq (transcriptome) data sets
- ☐ A web-based system for browsing genome data
- ☐ Elements of men's formal evening wear
- ☐ Programs for determining the function of a gene

7. Sequence alignment refers to

1 point

- ☒ Finding the best-matching position where a DNA sequence fits into a larger genome
- ☐ Finding the single nucleotide polymorphisms in a person's DNA
- ☐ Determining the amino acid sequence produced by translating a DNA sequence
- ☐ Shuffling the positions in a sequence to randomize them

8. A software pipeline for RNA sequence (RNA-seq) analysis will

1 point

- ☐ Create a database in which one can efficiently store and retrieve results
- ☐ Evaluate the best RNA-seq protocol for minimizing sequencing artifacts
- ☒ Process large raw sequence files into a summary table showing which genes were present
- ☐ Compare cases to controls and determine which genes were responsible for any differences in phenotype

9. Which of the following is not a computer operating system?

1 point

- ☐ Microsoft Windows
- ☐ RedHat Linux
- ☒ Google Drive
- ☐ Mac OS X

10. A data set large enough to overwhelm the main memory of a computer would

1 point

- ☒ Be larger than the available RAM
- ☐ Be at least 100 gigabytes in size
- ☐ Use uncompressed files rather than compressed ones
- ☐ Contain more than 1000 genomes' worth of data